Curriculum Vitae

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Yiwen Ding

PhD Candidate in Logic and Mathematics
Department of Ethics, Governance and Society
Vrije Universiteit Amsterdam, The Netherlands
Date of birth: 11 March 1997

dyiwen666@gmail.com
Website VU Profile
Google Scholar ResearchGate
0009-0000-6447-6479

Research Interests

My research connects category theory, algebra, logic in computer science, and artificial intelligence. I am particularly interested in using category theory to extend algebraic and logical theories from classical equational settings to ordered and quantitative settings, and in exploring their applications in computer science.

Education

- **PhD Candidate in Logic and Mathematics** Sep. 2022–present
  Vrije Universiteit Amsterdam, The Netherlands
  Supervisors: Alessandra Palmigiano and Alexander Kurz
  Doctoral thesis submitted; expected graduation by the end of January 2027.
- **MA in Philosophy** Sep. 2019–Jun. 2022
  Shandong University, China
  Advisor: Fei Liang
- **BA in Law** Sep. 2015–Jun. 2019
  Qufu Normal University, China
  Advisor: Yingkui Zhang

Doctoral Thesis

Quantale-Enriched Algebras

This thesis develops a unified framework for equational, ordered, and quantitative algebra using categories enriched over a commutative quantale. Its main results include a congruence–quotient correspondence, Birkhoff-style and Pavelka-style completeness for graded inequational logic, and enriched SP and HSP theorems characterizing classes of algebras definable by implications and inequations. These results also extend to separated algebras through separated reflection.

Publications

Journal Articles

1. Ruoding Wang, **Yiwen Ding** (corresponding author), Krishna Manoorkar, Apostolos Tzimoulis, Xiaolong Wang. Causal Kripke models. Synthese 206(3), Article 133, pp. 1–36; 2025. DOI: 10.1007/s11229-025-05223-z.

Refereed Conference Articles

1. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni, Apostolos Tzimoulis. Hybrid Logics with Moving Names. Dynamic Logic. New Trends and Applications (DaLí 2025); LNCS 16472, pp. 23–39; 2026. DOI: 10.1007/978-3-032-22626-6_2.
2. **Yiwen Ding**, Krishna Manoorkar, Apostolos Tzimoulis, Ruoding Wang. Probabilistic Causal Kripke Models. Logic, Rationality, and Interaction (LORI 2025); LNCS 16010, pp. 51–64; 2026. DOI: 10.1007/978-981-95-2481-5_4.

3. **Yiwen Ding**, Krishna Manoorkar, Alessandra Palmigiano, Ruoding Wang. Monotone Modal Logic Beyond Distributivity. Logic and Its Applications (ICLA 2025); LNCS 15402, pp. 111–125; 2025. DOI: 10.1007/978-3-031-89610-1_8.
4. **Yiwen Ding**, Krishna Manoorkar. Fuzzy Lattice-based Description Logic. Electronic Proceedings in Theoretical Computer Science 421, pp. 44–63; 2025. DOI: 10.4204/EPTCS.421.3.
5. **Yiwen Ding**, Krishna Manoorkar, Mattia Panettiere, Ruoding Wang. Toward the van Benthem Characterization Theorem for Non-Distributive Modal Logic. Advances in Modal Logic (AiML 2024); 2024. arXiv:2404.05574.
6. **Yiwen Ding**, Krishna Manoorkar, Apostolos Tzimoulis, Ruoding Wang, Xiaolong Wang. Causal Kripke Models. Electronic Proceedings in Theoretical Computer Science 379, pp. 185–200; 2023. DOI: 10.4204/EPTCS.379.16.

Conference Abstracts

1. **Yiwen Ding**, Alexander Kurz. Quantale-Enriched Universal Algebra: Completeness. Booklet of Abstracts, TACL 2026, p. 161; 2026. Abstract.
2. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni. Multi-type universal algebra: Transfer of properties. TACL 2024; Conference abstract. Abstract.
3. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni. Multi-type universal algebra: categorical equivalence. TACL 2024; Conference abstract. Abstract.

Preprints and Manuscripts

1. **Yiwen Ding**. Preservation and definability for the fluted fragment. 2026; under review. arXiv:2607.12970.
2. **Yiwen Ding**, Krishna Manoorkar, Ni Wayan Switrayni, Ruoding Wang. Defeasible Reasoning on Concepts. 2024; under review. arXiv:2409.04887.
3. **Yiwen Ding**, Krishna Manoorkar, Mattia Panettiere, Apostolos Tzimoulis, Ruoding Wang. Game semantics for lattice-based modal μ-calculus. 2023, revised 2024; under review. arXiv:2310.13944.

Formalization

Modal Characterization in Lean 4 2026

This project formalizes central results in classical modal model theory in Lean 4, building on mathlib and CSLib. It covers van Benthem's characterization theorem for basic unimodal logic, modal definability of pointed model classes, bisimulation-somewhere characterizations, and Hennessy–Milner theorems. Supporting constructions include saturation and ultraproducts.

Teaching Experience

- **Teaching Assistant — Logic, Information, Argumentation** Fall 2024
  Amsterdam University College, The Netherlands. Undergraduate course.
- **Teaching Assistant — Modal Logic** Fall 2021
  Shandong University, China. Undergraduate course.
  Course instructor: Chao Xu
- **Teaching Assistant — General Logic** Fall 2019
  Shandong University, China. Undergraduate course.
  Course instructor: Qinghui Su

Talks and Conference Presentations

1. Topology, Algebra, and Categories in Logic (TACL 2026). 2026. Abstract.
2. An Invitation to Quantale-Enriched Universal Algebra. GALAI Seminar, Chapman University, USA. Apr. 29, 2026. Abstract.

3. Dynamic Logic. New Trends and Applications (DaLí 2025). 2025. Paper abstract.
4. 7th Asian Workshop on Philosophical Logic (AWPL 2025). Kolkata, India. 2025. Programme.
5. ESSLLI 2024 Student Session. Leuven, Belgium. 2024. Programme.
6. Advances in Modal Logic (AiML 2024). Prague, Czech Republic. 2024. Abstract.
7. Topology, Algebra, and Categories in Logic (TACL 2024). Barcelona, Spain. 2024. Abstract.
8. MOSAIC Workshop 2023: Modalities in Substructural Logic II – Theory, Methods and Applications. TU Wien, Vienna, Austria. Sep. 27–29, 2023. Programme.

Academic Service

- **Journal reviewing:** Studia Logica; Fuzzy Sets and Systems; Annals of Pure and Applied Logic; Journal of Logical and Algebraic Methods in Programming.
- **Conference reviewing:** Advances in Modal Logic (AiML 2024, 2026); Dynamic Logic. New Trends and Applications (DaLí 2025); FedCSIS 2026.

Academic Visits

- Chapman University, USA
  Center of Excellence in Computation, Algebra, and Topology (CECAT)
  Mar. 2026; Oct. 2025 — Host: Alexander Kurz
  May 2024; Apr. 2023 — Host: Peter Jipsen
- University of the Witwatersrand, South Africa Jan. 2023
  School of Mathematics; Host: Willem Conradie

Scholarships and Support

China Scholarship Council: full funding for PhD studies in the Netherlands (2022–present).